Chemical Bonding: Comprehensive Lecture Notes

Page 1: Introduction to Chemical Bonding

Chemical bonding refers to the attractive forces that hold atoms or ions together to form more complex molecules and structures. In nature, most elements do not exist as isolated atoms; instead, they form bonds to achieve a state of lower potential energy and greater stability. The driving force behind this behavior is summarized by the octet rule.

The Octet Rule states that atoms tend to gain, lose, or share electrons until they are surrounded by eight valence electrons. This electronic configuration mimics that of the noble gases, which are exceptionally stable due to their filled outer s and p orbitals. Lewis symbols (or electron-dot symbols) are used to represent valence electrons as dots around the chemical symbol.

Page 2: Ionic Bonding and Lattice Structures

Ionic bonding occurs when valence electrons are completely transferred from a metal atom to a non-metal atom. This electron transfer creates positively charged cations and negatively charged anions. The electrostatic force of attraction between these oppositely charged ions constitutes the ionic bond.

A classic example is Sodium Chloride (NaCl). A neutral sodium atom loses one electron to form a sodium cation (Na+), while a neutral chlorine atom gains that electron to form a chloride anion (Cl-). These ions pack together into a highly organized, repeating three-dimensional network called a crystal lattice. The stability of this structure is measured by its lattice energy, which is the energy released when gas-phase ions coalesce into a solid lattice structure.

Page 3: Covalent Bonding and Molecular Compounds

Covalent bonding occurs when two or more non-metal atoms share valence electrons to achieve stable octets. Unlike ionic compounds, covalent molecules retain discrete molecular boundaries and exist as independent particles.

Covalent bonds can be single (sharing one pair of electrons), double (sharing two pairs of electrons, as in carbon dioxide), or triple (sharing three pairs of electrons, as in diatomic nitrogen). When atoms of different elements share electrons, the sharing is often unequal due to differences in electronegativity. Electronegativity is the relative ability of an atom in a molecule to attract shared electrons to itself. This unequal sharing creates polar covalent bonds.

Page 4: Metallic Bonding and the Electron Sea Model

Metallic bonding describes the electrostatic attraction between conduction electrons and positively charged metal ions. It is unique to metallic elements and alloys, and explains many of their distinct physical properties.

The electron sea model describes metals as a rigid lattice of metal cations immersed in a mobile, delocalized 'sea' of valence electrons. Because these conduction electrons are free to move throughout the entire structure, metals exhibit high electrical and thermal conductivity. Furthermore, this mobility allows the metal lattice to deform without breaking, explaining why metals are both malleable (can be hammered into sheets) and ductile (can be drawn into wires).

Page 5: Intermolecular Forces and Physical States

Intermolecular forces (IMFs) are the attractive forces that exist between separate molecules. While chemical bonds (covalent and ionic) hold atoms together within a molecule, IMFs hold molecules together in bulk solid or liquid states. IMFs are significantly weaker than chemical bonds.

There are three primary types of intermolecular forces: London dispersion forces (weak, temporary dipoles present in all molecules), dipole-dipole interactions (forces between permanent polar molecules), and hydrogen bonding. Hydrogen bonding is an exceptionally strong type of dipole-dipole interaction occurring when hydrogen is covalently bonded to highly electronegative atoms: Nitrogen, Oxygen, or Fluorine.